

LOAD CARRIAGE MODIFICATION

A PROJECT REPORT

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CHAPTER 1.

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INTRODUCTION

1. IDENTIFICATION OF CLIENT /NEED / RELEVANT CONTEMPORARY ISSUE :

Load carriage modification refers to the process of adapting or altering the way in which a person carries a load, such as a backpack or other type of bag, to optimize comfort, stability, and efficiency. This can be especially important for individuals who engage in activities that require extended periods of load carriage, such as hiking, military operations, or emergency response. When modifying load carriage, there are several specific points that should be considered to ensure optimal performance. These include the weight distribution of the load, the placement of straps and padding, the adjustment of the harness or frame, and the use of specialized equipment such as load-bearing vests or belts.

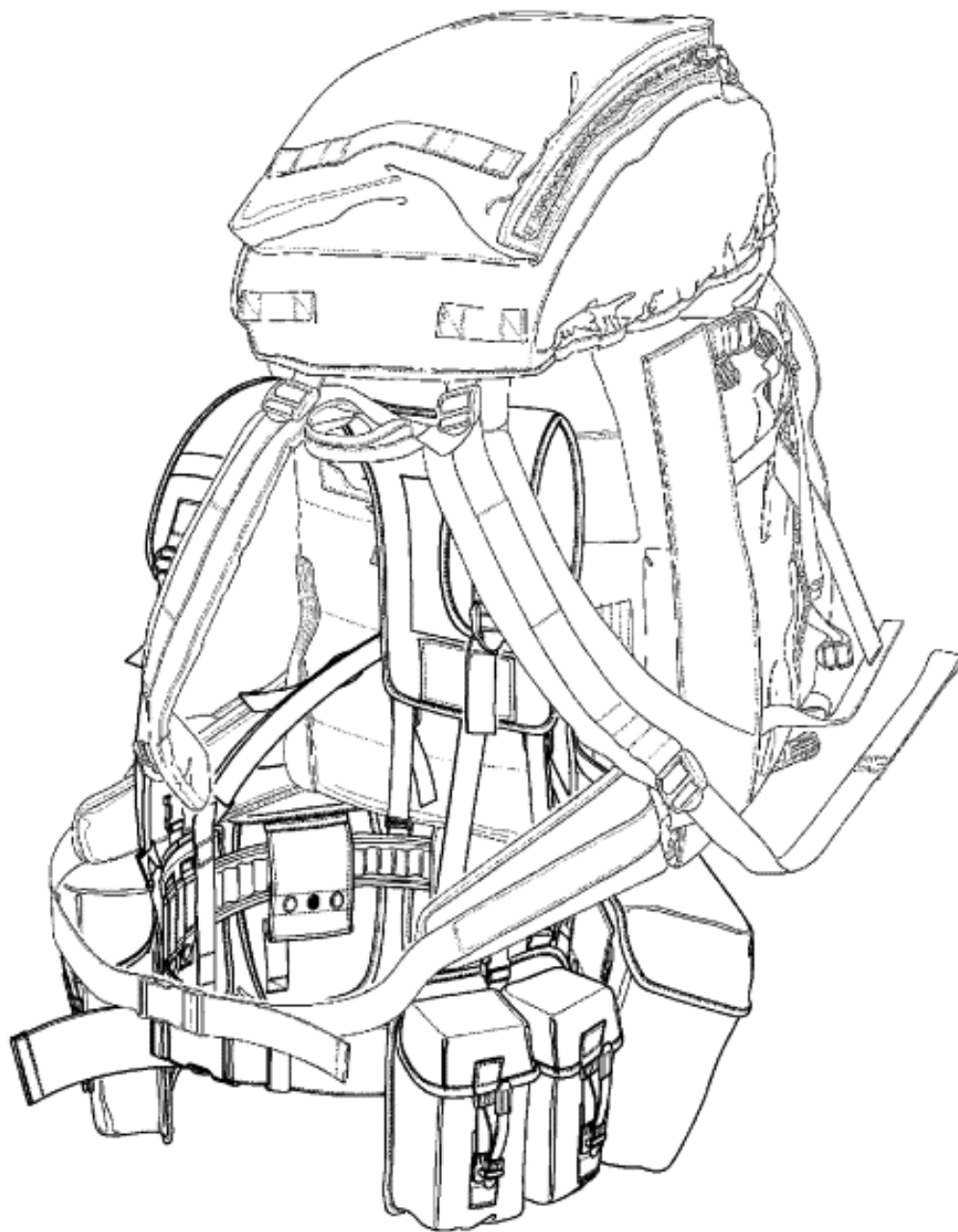
- **Cost Savings:** By optimizing load carriage efficiency, individuals can reduce the amount of time needed to complete tasks or activities, potentially resulting in cost savings in terms of labor and time.
- **Environmental Concerns:** To mitigate these environmental concerns, load carriage modification should prioritize lightweight and sustainable materials, as well as efficient design that minimizes the use of unnecessary components. Additionally, proper disposal of worn or damaged equipment should be considered to reduce the amount of waste generated from load carriage.
- **Compliance Requirements:** Load carriage may involve adjusting the height and position of the load, as well as the distribution of weight across the body. Load carriage systems should be designed to minimise the risk of accidents or injuries during use. Load carriage systems should be designed to distribute weight evenly across the body, to avoid placing excessive strain on any one area.
- **Planning and Forecasting:** To plan and forecast load carriage in a trolley, you should consider the following factors: Before loading any items onto the trolley, you should ensure that the weight of the items does not exceed the trolley's weight capacity. It's essential to distribute the weight of the items evenly across the trolley. It's important to consider the size and shape of the items you plan to load onto the trolley. Large or bulky items may require more space and can affect the stability of the trolley.

Once you have considered the above factors, you can plan and forecast load carriage in the trolley. It's recommended to load heavier items at the bottom of the trolley, ensuring that they are evenly distributed. Lighter items can then be loaded on top of the heavier items. You should also ensure that any fragile items are adequately protected to prevent damage during transport.

- **Businesses** -with the help of modern technology and various platforms to work on, modification to small but everyday items could help the flow of business to continue without affecting time consumption. A business that specializes in load carriage modification could offer various products and services related to improving the comfort, safety, and efficiency of carrying loads, such as backpacks, bags, and other types of gear.
- **Consumer product manufacturers** - Load carriage modification refers to the process of modifying existing consumer products to make them suitable for carrying heavy loads. This can include adding straps, buckles, and other attachments to backpacks, bags, and other types of luggage to make them more comfortable and secure when carrying heavy loads.
- **Utilities** - Due to load carriage modifications, versatility is introduced. Thus enhancing the overall experience of smart technology.



Load carriage special (figure:1.1)



Load carriage(figure:1.2)

2. IDENTIFICATION OF PROBLEM

The problem with the current load carriage and smart trolley is that the added technology such as the scanner add weights to the carriage cart. This reduces the efficiency and requires more force to utilise. Due to the nature of modification the cost of each cart varies depending on the required precision and efficiency. If more precision is required or is to be utilised, the efficiency of the overall product is reduced however if the efficiency is to be more preferred then the precision of the overall product is reduced.

In order to maintain the harmony between precision, efficiency and cost of the overall product, the raw materials used should be of better quality having to give out precise readings and less weight. Generally if these factors are to be taken into consideration, the manufacturing of these products will have to tested and approved by the GNP standards.

With recent investigation on such malls and market, the products are usually not packed definitively, thus making it difficult to work with.

These modification should also take into consideration of the power consumption. Since most of the added modification is to be functioned with power, mostly electricity, proper battery and capacitors are required. Since there is a thin line between reliability and the power supply, it is difficult to maintain harmony between it. For example, if the power consumption is high but the capacitance is low then the reliability of the product is reduced, however if the capacitance is high then the cost of the product is also high.



Functioned with power (figure:1.3)

3.IDENTIFICATION OF TASKS

- **Abstract:** This review focuses on the physiological factors that need to be taken into account when developing employment standards for jobs that heavily rely on load carriage. Heavy loads are frequently carried by workers in the military, firefighting, law enforcement, and search and rescue fields. Soldiers, for instance, frequently carry loads exceeding 50 kilograms, while structural firefighters also carry 20 to 25 kilograms of protective gear and clothing. Heavy loads are known to alter gait, mobility, metabolic rate, and efficiency, as well as increase the likelihood of muscle fatigue and injury. Additionally, protective gear adds to the workplace's thermal burden when load carriage occurs in environmentally stressful conditions.
- **introduction:** Both emergency service (public safety) and defense organizations have long had the expectation of finding and hiring a workforce that is highly capable and resistant to injury. However, there must be a balance between meeting those obligations and avoiding discriminatory hiring practices (Shephard, 1991; Jamnik and co. 2013; Tipton and co. 2013; Adams 2016), and with the understanding that employers should not be forced to make unreasonable or undue accommodations (Hatfield 2005; Canadian Commission on Human Rights in 2007; Adams 2016).
- **Historical considerations:** Rogers et al.'s terminology is used in this paper (2014) will be adhered to. To put it another way, an employment standard is a conceptual understanding of the characteristics recruits or incumbents must have in order to effectively and safely perform the essential work. The size of the carried loads is the first and most striking issue. Knapik et al., Lothian (1921a, 1921b, 1921c, and 1922), Historical evidence regarding the size of the loads carried by soldiers has been examined by Orr (2010) and 1996, 2004). A selection of those loads has been taken from the final source (Fig. 1), which weighed roughly 30 kg on average prior to the 1980s. Soldiers' capacity to carry loads increased in response to increasing levels of equipment sophistication and its mobilization. As a result, these loads have increased by more than 50% over the past 30 years, reaching an average of almost 50 kg across the three most recent international conflicts. Unfortunately, despite the fact that technological advancements have reduced the mass of some items that are carried and rendered obsolete others, the overall load that is carried has remained relatively constant. Instead, selection must be determined by the critical tasks' physiological demands. As a result, these load-carriage decisions have legal repercussions, which the following section explains in terms of their physiological basis.
- **Intensity dependency of endurance standards:** Beyond the scope of this contribution and its size, a comprehensive study of the effects of load carriage on physiological function is possible. Instead, readers are directed to historical works (such as Parkes 1866; 1901 Munson; Cathcart and co 1923), and Knapik et al.'s more recent manuscripts (Haisman, 1988) Stevenson et al. (1996), 2004) Epstein et al. (2004a), 2013). The results that are most likely to have an impact on thermal strain, cardiorespiratory function, and whole-body metabolic rate are the primary focus of this study. The creation of employment standards that are equitable and fair is directly impacted by these effects. Additionally, load carriage is common and frequently essential in physically demanding occupations, particularly those in public safety (such as police and firefighters).

Duty Position	Fighting Load (kg)	Approach March Load (kg)	Emergency Approach March Load (kg)
Rifleman	29	43	58
M203 Grenadier	32	48	62
Automatic Rifleman	36	50	64
Antitank Specialist	31	45	59
Rifle Team Leader	29	43	59
Rifle Squad Leader	28	43	58
Forward Observer	26	41	58
Forward Observer Radio/Telephone Operator	27	39	54
Weapons Squad Leader	28	45	60
M240 Machine Gunner	37	51	60
M240B Assistant Gunner	32	55	67
M240B Ammunition Bearer	31	53	65
Rifle Platoon Sergeant	28	41	54
Rifle Platoon Leader	28	42	53
Platoon Medic	25	42	54
Radio/Telephone Operator	29	45	No data
Mortar Section Leader	26	50	68
Mortar Squad Leader	28	58	65
60-mm Mortar Gunner	29	49	61
60-mm Mortar Assistant Gunner	25	55	No data
60-mm Mortar Ammunition Bearer	24	46	No data
Rifle Company Communication Chief	31	50	No data
Fire Support Officer	25	42	No data
Fire Support Noncommissioned Officer	24	41	65
Sapper Engineer	27	43	60
Company Executive Officer	27	42	No data
Company First Sergeant	29	41	57
Company Radio/Telephone Operator	29	44	59
Rifle Company Commander	30	44	50
Average	29	46	60

Table: 1

- **Physiological impact of load carriage:** At any intensity, load carriage unquestionably increases physiological strain and reduces the capacity for external work during near-maximal exercise. To put it another way, the body's added loads must be supported and moved with some of the available energy (or energy reserve). From a logical standpoint, the proportion of the available energy pool devoted to load carriage will be greater the greater the load's mass. As a result, there is less energy available for movement.
- **Metabolic cost:** Any task's tolerable work duration will be determined by the metabolic cost of that task and the intensity with which it is performed. These facts are applicable to all types of work, but they are frequently overlooked. An unacceptable bias will be introduced if that takes place in occupational research, which will ultimately be used to establish an endurance fitness cut-score (Tipton et al.). 2013), thereby opening the door to legal action (Jamnik et al. 2013).
- **Dependence of endurance standards on body size:** The matter of size dependency has a direct bearing on the development of employment screening tests and cut-scores, and a relevant example is found within the physiological aptitude testing of firefighters. It is appropriate to briefly reflect on endurance testing for occupations that routinely require load carriage. In many countries, the multi-stage, shuttle-run test (Léger and Lambert 1982) is used to evaluate the physiological endurance of candidates before entering various emergency services and the military. Whilst the prediction algorithms require modification for different applications (Stickland et al. 2003), the test provides a valid field prediction of peak aerobic power (Léger and Gadoury 1989; Wilkinson et al. 1999).

Table 3-9: Changes in Load Carriage Performance as a Function of Type of Training

Training Group	Pre-training	Post-training	Change (%)
Group 1 Total body resistance + aerobic	25:18	21:45	16
Group 2 Upper body resistance + aerobic	28:37	25:32	12
Group 3 Total body resistance only	29:27	28:12	4
Group 4 Aerobic only	30:32	30:31	0

Table: 2

4. TIMELINE:

Gantt Chart



Timeline Figure 1.4

5. ORGANIZATION OF THE REPORT

Chapter 1 Problem Identification: This chapter introduces the project and describes the problem statement discussed earlier in the report.

Chapter 2 Literature Review: This chapter presents review for various research papers which help us to understand the problem in a better way. It also defines what has been done to already solve the problem and what can be further done.

Chapter 3 Design Flow/ Process: This chapter presents the need and significance of the proposed work based on literature review. Proposed objectives and methodology are explained. This presents the relevance of the problem. It also represents logical and schematic plan to resolve the research problem.

Chapter 4 Result Analysis and Validation: This chapter explains various performance parameters used in implementation. Experimental results are shown in this chapter. It explains the meaning of the results and why they matter.

Chapter 5 Conclusion and future scope: This chapter concludes the results and explain the best method to perform this research to get the best results and define the future scope of study that explains the extent to which the research area will be explored in the work.

TEAM ROLES

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